Engineering Mathematics I - Calculus & Linear Algebra

Course: Engineering Mathematics I | Semester 1 | Contributor: Alex Chen

Comprehensive study manual covering eigenvalues, Cayley-Hamilton, partial derivatives and serie

Module 1: Matrix Theory and Linear Systems

* Rank of a matrix using echelon reduction and normal forms.
* Characteristic equations, Eigenvalues and Eigenvectors properties.
* Cayley-Hamilton theorem and application in computing inverse matrices.

Module 2: Differential Calculus & Taylor Expansions

* Eulers theorem on homogeneous functions and successive differentiation.
* Taylor and Maclaurin expansions for multi-variable functions.
* Maxima, minima, and Lagranges method of undetermined multipliers.

Module 3: Exam Tips & Formula Shortcuts

* Always verify trace = sum of eigenvalues and determinant = product of eigenvalues.
* For homogeneous equations AX = 0, non-trivial solutions exist if and only if detA = 0.

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